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Hybrid machine learning-RUSLE approach for soil erosion assessment in Aizawl district, Mizoram, Northeast India

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Abstract

Water-induced soil erosion is a critical driver of land degradation in the Eastern Himalayas, threatening agricultural productivity and regional stability. Previous assessments in Northeast India often rely on coarse global datasets or resource-intensive field approaches that lack the spatial granularity required for district-level management in complex terrains such as Aizawl. This study aimed to quantify the modelled average annual soil loss and delineate high-risk erosion zones using high-resolution remote sensing data. We employed a hybrid approach that integrated the Revised Universal Soil Loss Equation (RUSLE) with machine learning algorithms, specifically Random Forest (RF) and Support Vector (SV). Parallelized pixel-based operations for rainfall (R), Soil Erodibility (K), cover management (C), and conservation practice (P) factors were conducted in Google Earth Engine (GEE), whereas Slope length and steepness factor (LS) factor was computed in ArcGIS 10.8. The findings show Rainfall erosivity ranged from 850 to 1108 MJ mm ha^{-1} h^{-1} yr^{-1} and extreme topographic relief with LS values up to 83. The model results indicated that RF performed with higher accuracy ($R^2 = 0.93$) than SV ($R^2 = 0.44$) in estimating soil erosion. The estimated mean annual soil loss was 151.15 t ha^{-1} yr^{-1} using the RUSLE model and 149.64 t ha^{-1} yr^{-1} using RF. A significant positive correlation was observed between soil loss and the LS ($r = 0.34$, $p < 0.001$), the P factors ($r = 0.34$, $p < 0.001$) and the R factor ($r = 0.27$, $p < 0.001$), while soil loss was weakly and negatively correlated with K factor ($r = -0.03$, $p < 0.001$). Furthermore, the probability zonation map shows that more than 52% of the district falls under a slight erosion risk zone, while 45% falls under a moderate zone; the remains is shared between high-to very-high erosion-prone areas, primarily in barren lands, built-up areas, riverbanks, and steep terrains. These findings highlight a critical "erosion paradox" in which high forest cover masks localized degradation hotspots. These patterns identify critical hotspots where extreme topography overrides forest protection, allowing managers to prioritize high-risk riverbanks and steep slopes for targeted agroforestry and landslide prevention efforts. This study recommends the urgent need for LS-based zoning and the adoption of multi-strata agroforestry systems to mitigate sediment yields and landslides.

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Keywords Remote sensing and GIS, RUSLE model and machine learning, Soil erosion and land degradation, Eastern Himalayas

1 Introduction

Soil attributes are considered one of the most important resources that significantly contribute to the national economy and sustainable agricultural production [1, 2]. Globally, water-induced soil erosion poses a severe environmental threat, compromising food security and ecosystem services [3, 4], which has a direct or indirect effect on soil health [3]. Geographic Information Systems (GIS)-based soil erosion modelling predicts soil erosion patterns across landscapes [5] by integrating spatial data and erosion models to assess the risk and impact of soil erosion [6, 7]. Extensive agriculture, deforestation for farming practices [8], development activities, and mining lead to soil erosion, which negatively affects soil properties and crop productivity [9, 10]. Soil erosion also leads to the deterioration of water quality and sedimentation in lakes and rivers and an increased risk of flooding [11, 12]. Soil erosion resulting from deforestation and improper land management is considered one of the major global environmental problems that adversely affect forest resources, agriculture, water bodies, and ecosystem services [13–16]. Montanarella et al. [17] reported that the mean rates of soil erosion worldwide range between 12 and 15 t ha^{-1} yr^{-1} which results to an annual loss of 0.9 to 0.95 mm of soil from the surface [18]. From a management and protection perspective, areas susceptible to soil erosion must be accurately measured. The main issue with soil erosion research is the application of a field-based approach, which necessitates a controlled environment and many resources to determine the impact of soil erosion [19, 20].

Numerous models for estimating soil erosion have been categorized into empirical, semi-empirical, and process-based models. These models are summarized as the Universal Soil Loss Equation (USLE) [21], Revised Universal Soil Loss Equation (RUSLE) [22], hill slope model [23], European Soil Erosion Model (EUROSEM) [24], and other models [25]. The RUSLE is a more widely used model than the other models, mainly because of its dependency on several variables, such as physiography, precipitation, soil, land use practices, and conservation sustenance [26, 27]. By knowing the overall amount of soil loss in a given area, decision-makers can identify soil erosion-prone areas to manage soil-eroded areas sustainably [11]. Several studies have been conducted on soil erosion using the RUSLE model [11, 28, 29]. However, Bhattacharyya et al. [30] reported that the estimated amount of soil erosion in India is 147 million hectares, of which 46% is caused by precipitation. In the specific context of Northeast India, existing literature reviews report annual soil loss ranging from 229.5 t ha^{-1} yr^{-1} under jhum and up to approximately 836 t ha^{-1} yr^{-1} under other non-jhum land uses, such as agriculture, open forest, and wasteland [31]. Many studies have suggested that high slope length, monsoon intensity, and traditional farming, which mainly involves shifting cultivation (Jhum) are major causes of soil erosion [32, 33]. Recent studies in high-relief Himalayan environments have increasingly utilized geospatial technologies and morphometric analyses to prioritize watersheds for conservation [34]. For instance, integrated assessments of hydro-geological hazards in the north western Himalayas have demonstrated the efficacy of combining remote sensing with machine learning to identify sites susceptible to environmental stressors [35]. Furthermore, the integration of multiple influencing

factors within GIS has emerged as a robust tool for delineating environmental potential and risks in various Indian terrains [36]. Investigating the geomorphological and hydrological characteristics of sub-watersheds is now recognized as essential for predicting erosion potential and developing targeted preservation strategies [37].

In recent years, Machine learning algorithms (MLAs) such as Artificial Neural Networks (ANN), Random Forest (RF) and Support Vector Machines (SV) have become vital for environmental studies, offering superior spatial modelling and pattern extraction from complex geospatial data [38]. Although the RF model is popular, its accuracy can be limited by its dependence on specific training and testing datasets [38]. Consequently, researchers are shifting toward Deep Learning and Deep Neural Networks (DNN), which provide greater flexibility for handling intricate datasets. However, the Aizawl district in Mizoram remains underexplored despite ranking first among northeastern states in terms of land degradation rates, with approximately 0.18 million hectares affected [39, 40]. Despite its high forest cover, the region experiences frequent mass wasting and topsoil removal, leading to what is termed the "erosion paradox" [41]. The Aizawl district is among the most populated and fastest growing districts, with the majority of the population dependent on farming practices (Jhum cultivation) [42]. Several studies have been conducted on soil, diversity, and land suitability aspects [41, 43–45] while limited information is available on soil erosion in the region. Aizawl is characterized by high undulating topography, steep slope terrains, and deep valleys. This rough terrain, composed of unconsolidated sedimentary deposits, slope failures, and heavy rainfall, often causes soil erosion and large landslides [46]. Therefore, this study addresses the local research gap of how slopes, cover management, and rainfall can lead to soil erosion by integrating Climate Hazards Center InfraRed Precipitation with Station data (CHIRPS), Sentinel-2 imagery, and Dynamic World data within a cloud-based geospatial environment to assess soil erosion and severity classes using the RUSLE model and Machine Learning (ML). While traditional RUSLE relies on a static multiplicative formula, the RUSLE-ML approach (specifically Random Forest, and Support Vector) provides higher predictive accuracy by capturing complex, nonlinear interactions. This machine learning integration enables the processing of high-resolution satellite data to better identify localized windows of vulnerability.

Our novelty and contributions are the development of a hybrid RUSLE-ML workflow in GEE that maintains technical pression in extremely rugged terrain, which will help in monitoring high-relief catchments [35–37]. The study's findings provide insights into the drivers of soil erosion, helping to build and implement effective management solutions to address this environmental issue.

This paper is structured as follows: Sect. 2 describes the study area, datasets, and the integrated RUSLE–machine learning methodology employed. Section 3 presents the results of rainfall erosivity, topographic, land cover, and conservation practice factors, along with soil erosion estimates and model performance comparisons. Section 4 discusses the key drivers of soil erosion in the context of regional geomorphology and land management practices. Finally, Sect. 5 summarizes the main conclusions, management implications, and limitations of the study.

2 Materials and methods

2.1 Study area

Geographically, Aizawl district, Mizoram exhibits variations in altitudinal ranges from 700 to 1288 m above sea level (a.s.l) and is located on a north–south ridgeline (Fig. 1). It is characterized by four distinct seasons: a cool dry winter (December to February) and a transition during the retreating monsoon (October to November). The mean annual precipitation in Aizawl area is reported to be 2,100–2,400 mm. The general soil texture is clay loam with a low pH (approximately 4.6–6.1) [47]. Aizawl city is the most densely populated in Mizoram, with about 78.6% of the district's residents in urban areas in 2011 [48]. It serves as the state's administrative, commercial, and cultural center, which places significant development and environmental pressure on its fragile ecosystem. The area is covered by diverse forest types, mainly subtropical broad-leaved evergreen forests interspersed with extensive bamboo, but is exposed to various anthropogenic pressures, including shifting cultivation and urbanization. The area's combination of undulating topography, high annual precipitation, soil's porosity, permeability, decrease in forest cover, and shifting cultivation practices, makes the area exceptionally prone to erosion and landslides [42, 49].

2.2 Integrated geospatial modeling framework

Soil loss was estimated using RUSLE model and Machine Learning (ML) (Eq. 1) [22] and Google Earth Engine (GEE) platform, which provides a vast library of satellite and climate data, making it time- and cost-effective compared to the old method of downloading single tiles from the USGS for further processing in GIS interface software.

$$A = R \times K \times LS \times C \times P. \tag{1}$$

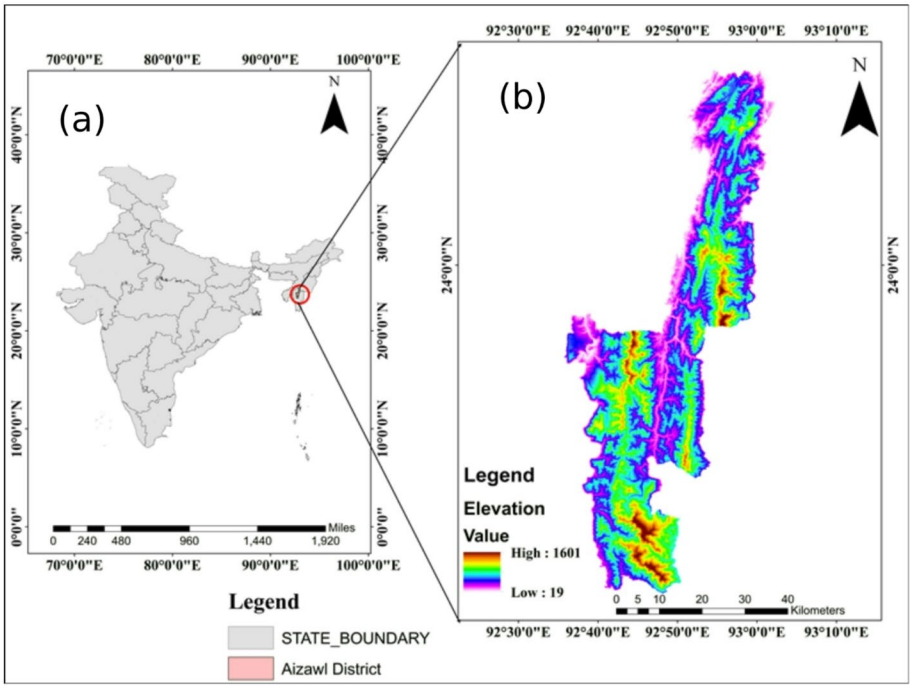


Fig. 1 Study area: **a** India, and **b** Aizawl district and elevation in meter (m)

Where, A: spatial average soil loss (t ha^{-1} yr^{-1}), R: rainfall erosivity factor (MJ mm ha^{-1} h^{-1} yr^{-1}), K: soil erodibility factor (t ha h $ha^{-1}MJ^{-1}$ mm^{-1}), LS: slope steepness and length (dimensionless), C: cover management factor (dimensionless, ranging from 0 to 0.5), and P: supporting conservation practice factor (dimensionless, ranging between 0 and 1). To ensure methodological transparency, this study adopted a hybrid processing approach. While R, K, C, and P factors were processed using parallel pixel-based operations in GEE, the LS factor was computed in ArcGIS 10.8. This decision was necessitated by the requirement for recursive neighbor-to-neighbor pixel calculations (e.g., D8 flow accumulation) in steep terrain, which are more computationally stable in desktop GIS environments [15]. The detailed methodology flowchart is presented in (Fig. 2).

2.3 Rainfall erosivity factor (R)

Rainfall is the primary component of the RUSLE model for estimating soil erosion. The R factor was examined following Wischeier and Smith [21] using the total storm energy and maximum 30-minute intensity. Because of the absence of long-term ground rain-gauge rainfall data for the Aizawl district, the rainfall erosivity factor was calculated for the period of 2001–2024 using daily grided rainfall data from Climate Hazards Group InfraRed Precipitation with Station data (CHIRPS) (https://www.chc.ucsb.edu/data/chirps) at 0.05° resolution. To calculate R, daily CHIRPS data for the period 2001–2024 were first aggregated into annual sums to derive the long-term mean annual precipitation (P) using functions in GEE and then exported to drive to visualize the final image in the results. We generated random points to extract values from the rainfall grids and then

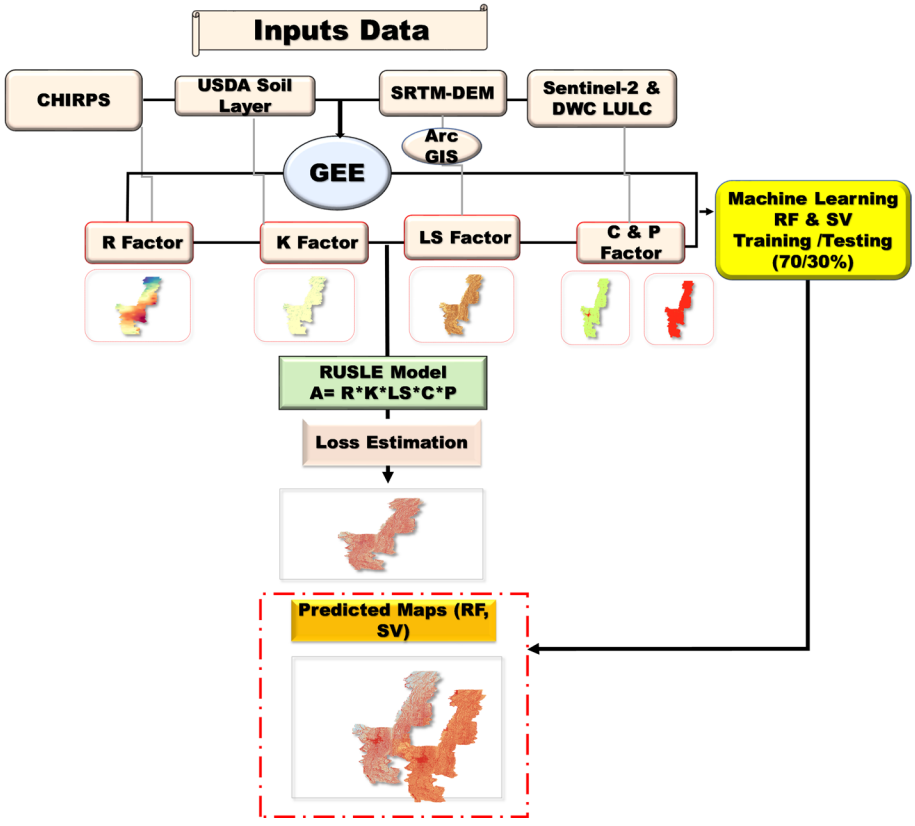


Fig. 2 Integrated RUSLE and Machine Learning (ML) analytical framework

applied the Inverse Distance Weighted (IDW) interpolation method to the grided in ArcGIS 10.8 using a power parameter of 2.0 to preserve localized orographic influences and smoothening the irregular boundaries. Although CHIRPS is a satellite product, regional validation studies in Northeast India have shown that it maintains a strong correlation ($R > 0.80$) with station data [50], although it may slightly underestimate extreme daily precipitation peaks in high-altitude zones [51]. The relationship was established using an equation optimized for Indian conditions (Eq. 2) [52].

$$R = 79 + 0.363 * P. \tag{2}$$

where, R represents the rainfall erosivity factor (MJ mm ha^{-1} h^{-1} yr^{-1}), and P: mean annual rainfall (mm).

2.4 Soil erodibility factor (K)

The soil erodibility factor (K) represents soil vulnerability to erosion and influences different soil properties [21, 53]. Soil data were obtained from the United States Department of Agriculture (USDA; https://g.co/kgs/TWHbv8a) soil classification database (250 m resolution), which provides texture and organic matter classifications that were used to drive the K-factor values in this study. Soil classes are associated with K values ranging from 0 to 1, with higher values indicating greater susceptibility to soil erosion. These values are significantly different among land types and can be estimated using soil texture and its corresponding organic matter content as per the soil erodibility factor [54]. We used soil texture of at 0 depth centimetres band (0b) to generate texture map and K values.

2.5 Slope length and steepness factor (LS)

The slope length factor (LS) quantifies the effect of slope length and steepness on erosion. However, a high slope length and gradient have a more erosive power of runoff. To compute the LS factor, the Shuttle Radar Topography Mission (SRTM) from satellite data with a spatial resolution (30 m) was utilized to generate a Digital Elevation Model (DEM) and flow accumulation, and then the LS factor was calculated using Eq. (3). This step was performed using Arc Hydro tools in ArcGIS desktop 10.8, because GEE requires recursive pixel-to-pixel calculation (each cell depends on upstream neighbours) and is not built for those iterative spatial operations, rather it is for parallel per-pixel processing. The technical steps included: (1) filling artificial sinks to ensure hydrological continuity, (2) applying the D8 flow direction algorithm to determine the path of steepest descent, and (3) calculating flow accumulation using the default threshold (1%).

$$LS = [(Flow\,accumulation) * Cellsize/22.13]0.4 * (sin)\,(\theta/0.0896)1.3 \tag{3}$$

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2.6 Cover management factor (C)

The Cover management factor (C) represents the impact of cover and management strategies on soil erosion. The values of the C factor range from 0 to 1 (no cover to bare ground) and it was computed using Sentinel-2 NDVI using the formula described by Van der Knijff et al. [55] Eq. (4).

Table 1 Conservation practice factor (P) values across different land use

LULC Type	Slope % range	*P*-Factor value
1- Croplands	0–5	0.10
	5–15	0.12
	15–30	0.14
	30–50	0.19
	50–75	0.25
	> 75	0.33
2- Other Land use		1.00

Table 2 Summary of dataset used

Dataset Name	Data provider	Date range/mean
Sentinel-2 L2A	Copernicus / ESA	January- June – 2024 with (5%) cloud cover
CHIRPS Precipitation	Climate Hazards Center (CHPC)	2001–2024
Digital Elevation Model (DEM)	USGS	
Dynamic World Land Cover	Google / World Resources Institute	January- June – 2024 with (5%) cloud cover

$$C = expr[-\alpha * \left(\frac{NDVI}{\beta - NDVI}\right)]. \quad (4)$$

Where, α = 1 and β = 2 are parameters that determine the NDVI-C curve shape. The Values of (β) and (α) are based on [55].

2.7 Conservation practice factor (P)

Conservation practice factor (P) values and map was generated by integrating dynamic world cover data near real time (NRT) (https://dynamicworld.app/), having 10 m resolu tion, derived from the Sentinel-2 L C, land use/land cover (LULC) dataset in GEE. This dataset includes class probabilities and label information for nine classes, with an overall accuracy of approximately 82% for experts and 73% for non-experts in the true class [56]. This process produced P-factor values and a map exhibiting an overall accuracy of 76.7%. The P values were subsequently allocated according to land use type, as specified in the USDA manual [57, 58] (Table 1). The summary of the dataset used in this study is presented in Table 2.

2.8 Weighted Index Overlay (WIO) probability

To determine the potential risk zones, a Weighted Index Overlay (WIO) is performed. This method assigns specific weights to three environmental drivers: Land Use (40%), Slope (35%), and Rainfall (25%). The resulting index identifies four probability zones (low to very high) for future erosion management following Ganasri and Ramesh [59].

3 Results

3.1 Rainfall erosivity and soil erodibility factors

The findings revealed that the mean annual rainfall ranged from 2,124 to 2,822 mm per year (Fig. 3a). The value of R factor ranged from 850 to 1,108 MJ mm ha^{-1} h^{-1} yr^{-1} with a mean value of 979 (Fig. 3b). The north western parts of Aizawl receive the most rain, whereas the eastern parts of the region receive slightly less precipitation (Fig. 3a, b). It was also observed that the area is mostly dominated by clay loam (Fig. 3c). The soil

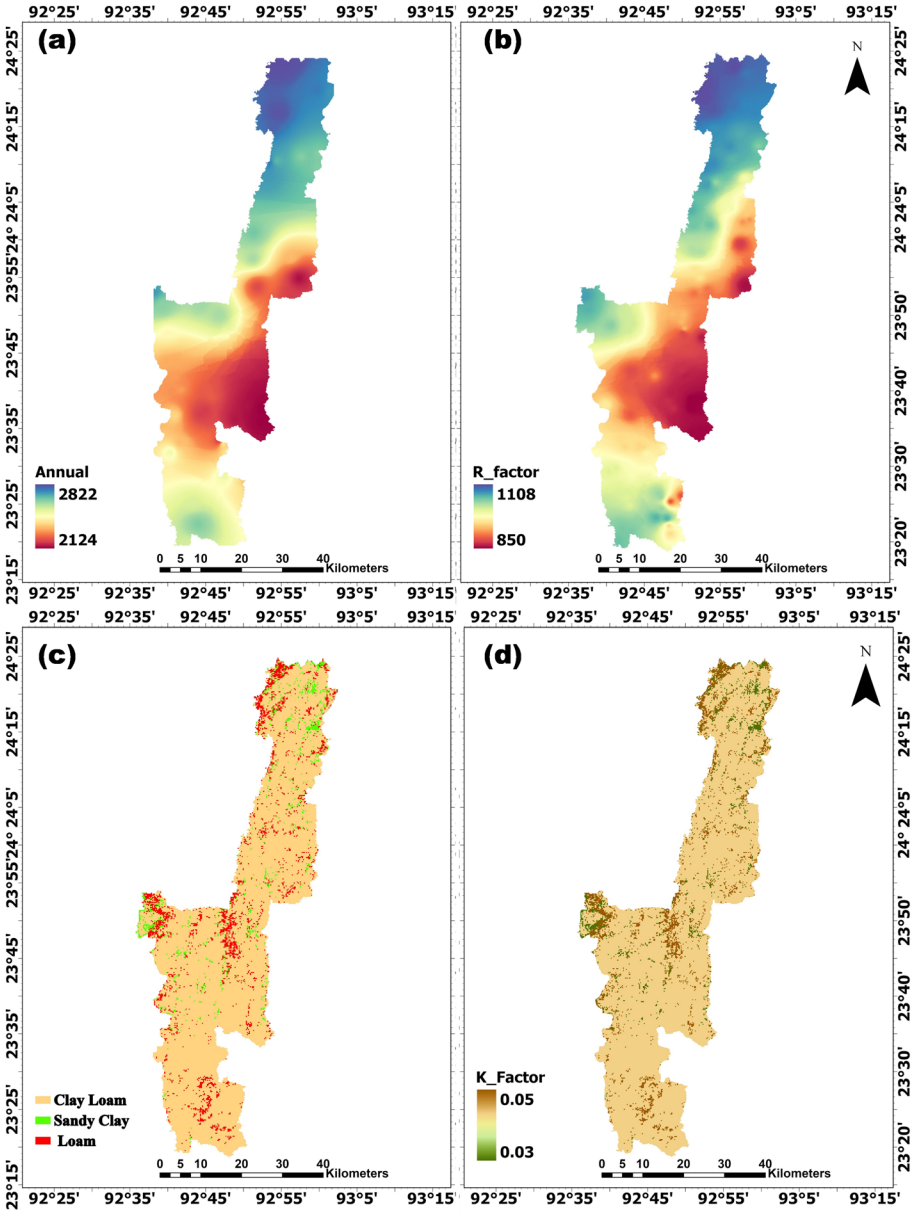


Fig. 3 Spatial distribution of rainfall and soil-related erosion factors in Aizawl district: **a** mean annual rainfall (mm) derived from CHIRPS data (2001–2024), **b** corresponding rainfall erosivity factor (R; MJ mm ha^{-1} h^{-1} yr^{-1}), **c** dominant soil texture classes, and **d** estimated soil erodibility factor (K; t ha h ha^{-1} MJ^{-1} mm^{-1}), illustrating relative uniform soil susceptibility across the district

erodibility factor (K) values ranged between 0.034 and 0.05 Mg ha^{-1} MJ^{-1} mm^{-1} with a mean value of 0.04 Mg ha^{-1} MJ^{-1} mm^{-1} (Fig. 3d).

3.2 Slope-steepness and length factor (LS)

The slope length, which quantifies the effect of slope length and gradient on soil erosion ranged from 0 to 83 across the study area (Fig. 4a), with a zero-degree indicating a flat area as observed in the Lengpui airport site (middle-left of Fig. 4a), while a very steep slope was observed in most of the mounting cliffs and terrain of the study area, which is associated with higher sediment transport.

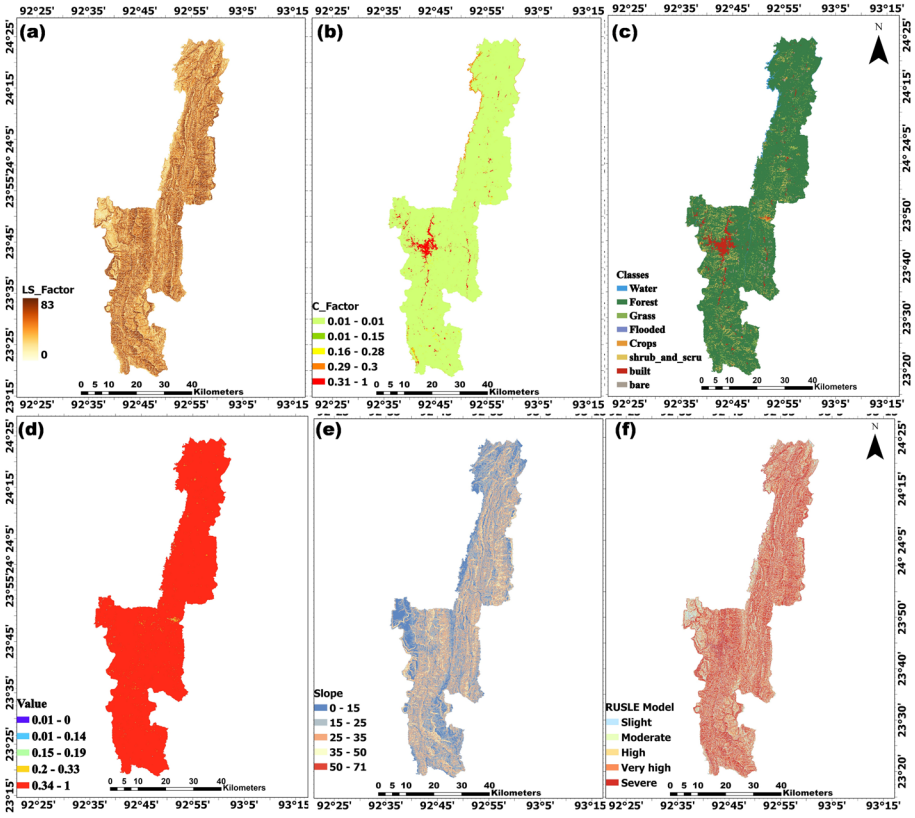


Fig. 4 Spatial pattern of key RUSLE factors and erosion severity in Aizawl district: **a** Slope-steepness and length factor (LS) (degrees), highlighting areas of extreme topographic influence, **b** cover and management factor (C) derived from Sentinel-2 NDVI, **c** Land Use-Land Cover (LULC) from dynamic world classification, **d** conservation practice factor (P) reflecting land management conditions, **e** slope (degrees), and **f** soil erosion severity classes (t ha^{-1} yr^{-1}) estimated using the RUSLE model

3.3 Cover management factor (C) and land use/ cover

The C factor represents the influence of land cover and management practices on soil erosion control, where lower values of C indicate greater erosion resistance. The C factor values varied from 0.01 to 1.0 (Fig. 4b). The C factor values for forest cover ranged between 0 and 0.28 (good cover), while the values for barren land, built-up land, and water bodies were close to 1 (Fig. 4c). This highlights that the potential degree of soil erosion is significantly different for different land-use types.

3.4 Conservation practice factor (P)

Support practice (P) can control or reduce soil erosion potential and the P values ranged between 0.01 and 1. The barren land, riverbank, possibly cropland, and settlement areas showed higher P values (Fig. 4d) with steeper slopes (Fig. 4e). Lower P values represent high support practice on soil, while lower values represent lesser or unavailability of conservation practices.

3.5 Estimation of annual soil loss according to RUSLE

To assess the severity of soil erosion, the study area was categorized into five classes based on the amount of soil loss per pixel: slight (< 10 t ha^{-1} yr^{-1}), moderate (10 to 20 t ha^{-1} yr^{-1}), high (20 to 30 t ha^{-1} yr^{-1}), very high (30 to 40 t ha^{-1} yr^{-1}), and severe (> 40 t ha^{-1} yr^{-1}), following the guidelines by Das et al. [60]. The spatial distribution of the

Table 3 Annual soil loss classes

Erosion class	Area (%)		
	RUSLE	RF model	SV model
Slight	27.91	27.57	23.1
Moderate	1.65	3.5	4.68
High	6.17	6.28	9.81
Very high	21.14	20.47	23.67
Severe	42.9	41.96	38.51

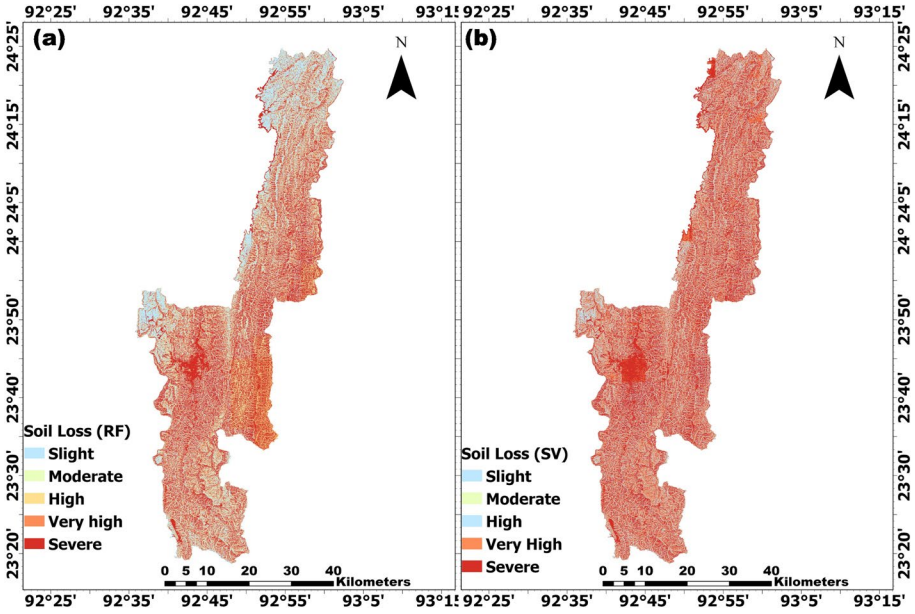


Fig. 5 Machin learning annual soil loss modelling: **a** Random Forest (RF), and **b** support Vector (SV)

RUSLE baseline was observed across nearly all erosion classes. The random forest model currently demonstrates the highest performance alignment with baseline (Table 3). For instance, in the slight erosion class, the RF model predicted 27.57%, which was nearly identical to the RUSLE estimate of 27.91%. Similarly, in the severe category, the RF model's estimate of 41.96% was very close to the baseline of 42.9%. Although the SV model also shows a similar trend, it remains slightly more divergent. It estimated a higher percentage of land in the high (9.81%) and very high (23.67%) categories than both RUSLE and RF. The RUSLE and RF-based models showed substantially equal results in terms of erosion severity categorization, where they assessed the annual average of soil erosion at 151.15 t ha^{-1} yr^{-1} (Fig. 4e) and 149.64 t ha^{-1} yr^{-1} (Fig. 5a), respectively (Fig. 5b). In contrast, SV showed a significant difference, with a mean of 36 t ha^{-1} yr^{-1}. Based on the correlation (Fig. 6), we found that the RF model demonstrated superior accuracy with an R^2 = 0.93, showing good alignment with the RUSLE baseline across all erosion classes. In contrast, the SV model performed poorly with an R^2 of 0.44, indicating underestimation of soil loss as erosion intensity increased. These values are higher than those reported by [58, 61], who reported a mean annual soil loss of more than 23-25.4 t ha^{-1} yr^{-1}. This could be attributed to the differences in land use practices and datasets used among the study regions. For example, both studies used 30 m resolution Landsat and global land cover (GLC) data for erosion control practices (P factor), whereas our study used 10 m dynamic world land use data. A significant positive correlation was observed between

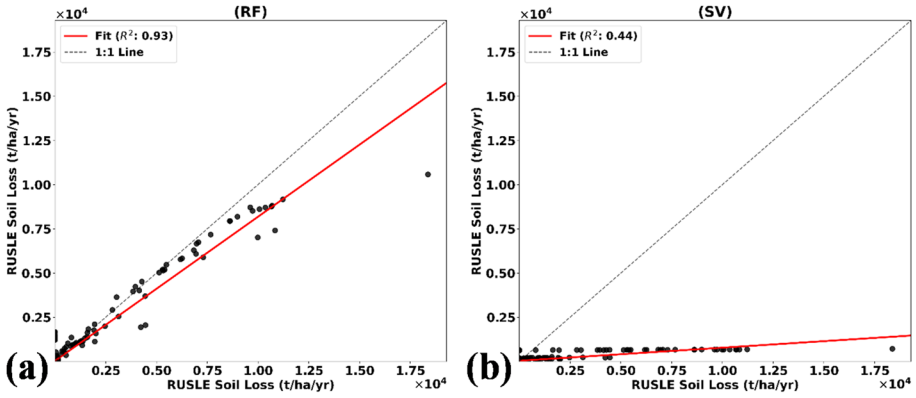


Fig. 6 Relationship between RUSLE-estimated soil loss and machine learning predictions: scatter plots showing the model performance for **a** Random Forest (RF) and **b** Support Vector (SV), including the 1:1 reference line and coefficient of determination (R^2), indicating stronger agreement between RF predictions and RUSLE baseline values

soil loss and the LS (r = 0.34, p < 0.001), the P factors (r = 0.34, p < 0.001) and the R factor (r = 0.27, p < 0.001), while soil loss was weakly and negatively correlated with the K factor (r = -0.03, p < 0.001) (Fig. 7).

4 Discussion

The Rainfall Erosivity (R factor), which ranged from 850 to 1108 MJ mm ha^{-1} h^{-1} yr^{-1}, aligns with the findings of Handique et al. [58] who reported R factors of (1,110.94 to 943.98 MJmmha^{-1} h^{-1} yr^{-1}) in Majuli Island, Assam, and other regions [62–64]. This high erosivity is influenced by the Southwest Monsoon from the Bay of Bengal, which receives precipitation with exceptionally high kinetic energy, which is one of the driving agents for the detachment of soil particles initiating the erosion process. This energy is then triggered by the topography (LS factor), which ranges up to 83 (Fig. 4a) and indicates highly undulating terrain with very long steep gradients; the lowest steeps are visible in flat surfaces, such as the Lengpui Airport site, while long steep is in mountain ranges and river banks. These values are in same range of findings found by Handique et al. [58] corroborating findings from Parveen and Kumar [14], Sheikh et al. [65], and Sud et al. [66]. Ran et al. [67] similarly highlighted the profound influence of slope steepness and length on rainfall runoff and soil erosion, highlighting how this powerful physical environment creates severe sediment transport channels. This also aligns with the national mapping by Raj et al. [68] who identified Northeast India as one of the most erosive regions in the country. Furthermore, projections by Das and Sarma [69] suggest that the intensification of monsoon events under climate change will likely increase regional R-factors by up to 27% under the SSP245 scenarios. While soil erosion is a complex process influenced by properties such as organic material, porosity, and texture [70]. The calculated K-factor values ranged from 0.03 to 0.05 t ha h $ha^{-1}MJ^{-1}$ mm^{-1}, indicating a narrow but consistently moderate to high susceptibility. This sensitivity is higher than the national mean of 0.028, largely due to the low organic matter and acidic nature of the regional clay loams [42, 68]. This value is consistent with those reported by Elnashar et al. [71] and Kumar et al. [72].

The final calculated soil loss was generated through highly localized management issues, particularly evident in the Cover Management (C factor), which ranged between

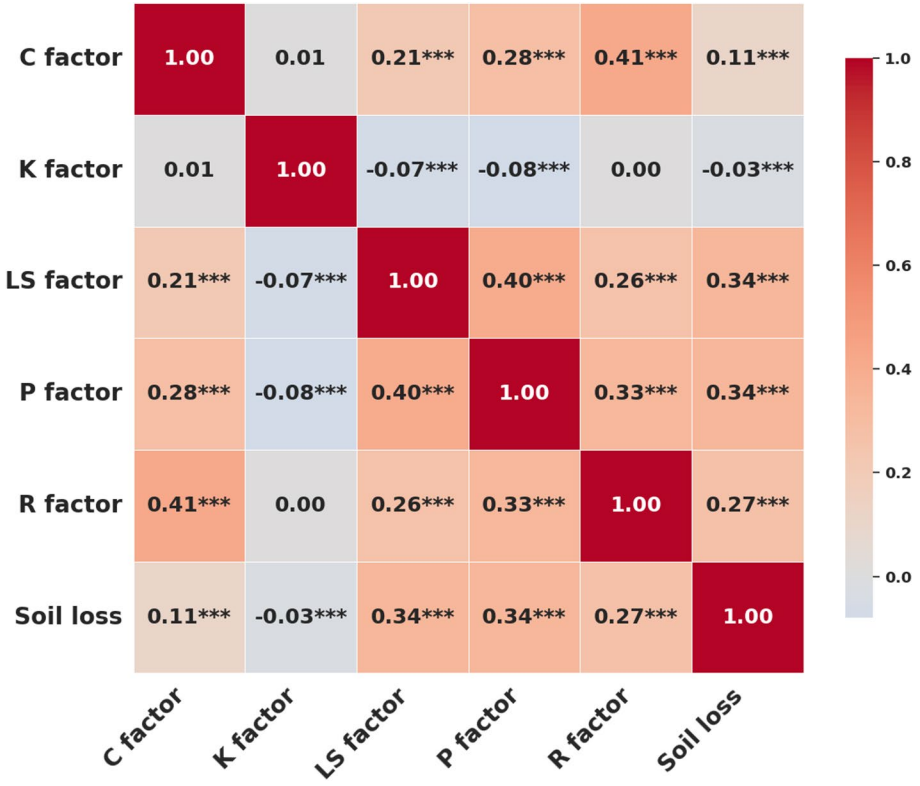


Fig. 7 Correlation matrix between RUSLE factors and annual soil loss in Aizawl district. Asterisks (***) denote statistical significance at the $p < 0.001$ level

0.01 and 1.0, consistent with Das et al. [60] and Dutta et al. [73]. This factor is dependent on land use type, with lower values indicating protective vegetation cover and higher values indicating barren, cropland or built-up areas. This leads to a critical Erosion Paradox: despite Mizoram being among the top five Indian states in terms of forest cover [74], the region experiences high annual soil loss.

Correlation analysis revealed that soil erosion is primarily driven by the region's extreme topography and anthropogenic land management rather than by inherent soil characteristics (Fig. 7). The LS factor ($r = 0.34$, $p < 0.001$) and P factor ($r = 0.34$, $p < 0.001$) exhibited the strongest positive correlations with soil loss, confirming that the district is characterized by steep slope gradients and a localized lack of structural conservation measures such as terracing or contouring, which are critical determinants of land degradation. Our findings reported a lower steep length than that reported by Barman and Das [41], which could be due to the differences in resolution between the SRTM (30 m) and The Cartosat-1 Digital Elevation Model (Carto DEM) with a 2.5 and 10 m resolution. Intense monsoonal rainfall (R factor, $r = 0.27$, $p < 0.001$) served as a significant climatic trigger for sediment transport, whereas the a weak and negative correlation of the K factor ($r = -0.03$, $p < 0.001$) indicated that soil erodibility was relatively uniform. Furthermore, the strong inter-correlation between the C and R factors ($r = 0.42$, $p < 0.001$) underscores that the impact of heavy rainfall is exacerbated by land cover management. This relationship provides a quantitative explanation for the erosion paradox, in which high forest cover persists alongside high sediment yields [75]. This paradox is resolved by the dominance of geomorphological factors over biological buffers in high-energy

monsoonal systems. In the meantime, rainfall characteristics often exert a more profound influence on erosion than land cover density alone [33]. In Aizawl, this is intensified by "windows of vulnerability" created by shifting cultivation, where pre-monsoon clearing leaves the soil exposed, riverbank sand quarrying, and the presence of high-cliff terrain [69].

Our modelled mean soil loss (151.15 t $ha^{-1}yr^{-1}$) is consistent with that reported for the Upper Tuirial Basin (115.4 t $ha^{-1}yr^{-1}$) by Kumar et al. [62] and the West Kameng Watershed (124.21 t ha^{-1} yr^{-1}) studied by Das et al. [60]. The higher district-wide values compared to the Chite watershed (6.10 t ha^{-1} yr^{-1}) could be attributed to the inclusion of extreme ridge-valley gradients located outside the urban catchment [76] and the use of the RUSLE-ML based approach. Similar dynamics have been reported by Maurya et al. [77] in the Beas Valley, where high topographic relief acts as a force multiplier for erosion, even under moderate rainfall [78–80]. Based on the overlay weightage analysis erosion suitability map (Fig. 8), we found that 52.2% and 45% of the total district area fell under the slight and moderate erosion zones, respectively. Most of the erosion occurs in riverbanks, road sides, and residential areas, which lack dense vegetation or are influenced by urban runoff and are highly susceptible to extreme soil loss. This was further validated by existing landslide data [41, 81, 82] and field-validated images (Fig. 9a-e). A summary of the statistical correlations and factor values is presented in Table 4.

5 Limitations and future research direction

While this study offers key insights into soil erosion in the Aizawl district, the accuracy of the RUSLE model and machine learning predictions depends on the availability and quality of input data, including rainfall, soil properties, and land cover, particularly historical rainfall records and satellite data. Despite its strong performance, the hybrid RUSLE–machine learning framework remains limited in capturing process-based erosion mechanisms such as landslides and deep gully formation, highlighting the need for future integration with physically based hydrological and slope stability models.

Therefore, these results should be used as general management guidelines rather than exact measurements. On the other hand, uncertainty in the results arises from the spatial resolution of the DEM used for LS factor derivation, the sensitivity of LS to slope and flow accumulation in steep terrain, potential underrepresentation of short-duration extreme rainfall events in gridded precipitation data, and the empirical nature of RUSLE, which does not explicitly represent gully erosion, landslides, or sediment transport processes.

Future research should focus on gathering better on-the-ground data and combining erosion estimates with slope stability factors to improve the prediction of long-term risks such as landslides. Moreover, future studies should focus on integrating higher-resolution topographic data, incorporating extreme rainfall dynamics, and coupling RUSLE-based machine learning frameworks with process-based hydrological and slope-stability models to better capture erosion mechanisms in steep, monsoon-dominated landscapes.

6 Conclusions and recommendations

In this study, we utilized the Revised Universal Soil Loss Equation (RUSLE) model, integrating machine learning and high satellite resolution data to estimate the annual soil loss in the Aizawl District. The model calculates erosion by combining five key parameters:

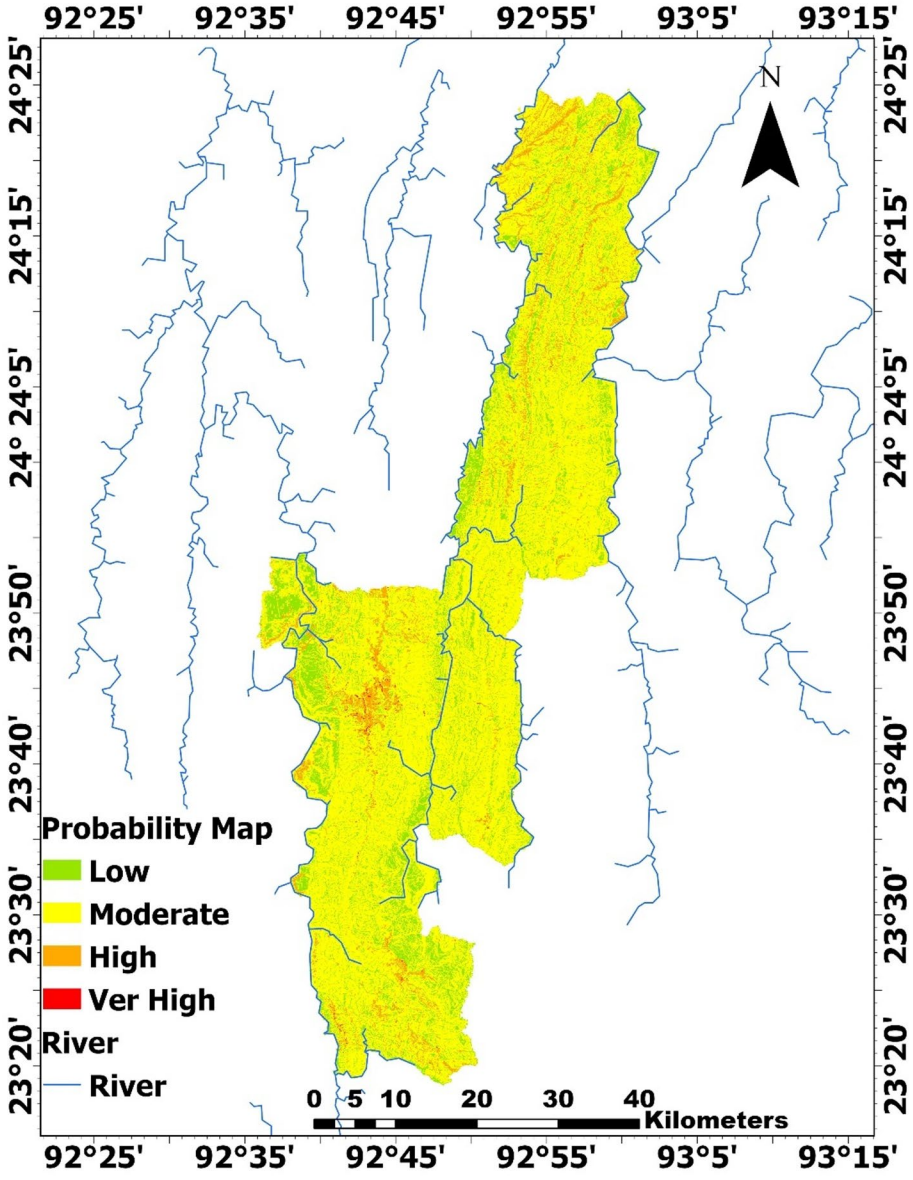


Fig. 8 Soil erosion probability map

rainfall erosivity (R), soil erodibility (K), slope length and steepness (LS), cover management (C), and support practice (P). The study revealed that the random forest model performed with higher accuracy than the support vector model (Fig. 6) compared to the baseline RUSLE model. The statistical correlation analysis among RUSLE parameters indicated that the length and steepness factor had a high impact on erosion (Fig. 7), followed by the management practice factor. Conversely, the K factor showed a moderate correlation with soil loss, but erosion was triggered by monsoon rainfall. The high erosion rates observed require prompt and well-coordinated erosion control measures by transitioning Jhum sites to multi-strata agroforestry using species such as Areca nut, Orange, and Pineapple, which can reduce effective C-factors, as stated by Obiefuna and Asoegwu [83]. In the Himalayan region, Maurya et al. [84] recommended integrating scientific soil conservation methods with indigenous practices, such as pre-monsoon plowing, filter strips, earthing-up, mulching, and soil compression, to enhance erosion

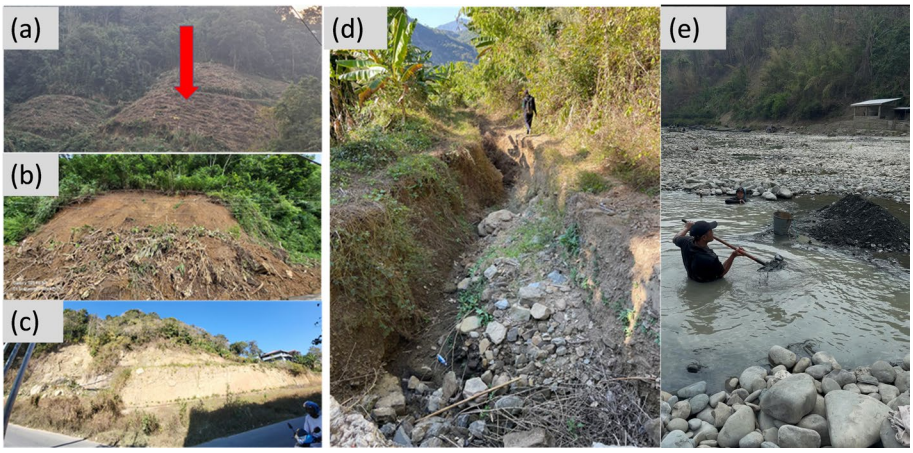


Fig. 9 **a** Forest clearing for "*Jhum*" cultivation in western part of Aizawl city, **b** excessive rainfall leading to soil erosion and eventually landslide, **c** land excavation activities mainly for construction purposes, **d** steep slopes erosion and **e** sand excavation from Tlawng river riverbanks

Table 4 Statistical summary and distribution and correlation of RUSLE factors

Factor	Range/distribution	Correlation with soil loss (*r*)
R	850–1108 MJ mm ha^{-1} $h^{-1}yr^{-1}$	0.27***
K	0.034–0.05 t h / MJ / mm	−0.03***
LS	0–83	0.34***
C	0.01–1.0	0.11***
P	Localized (Conservation absence)	0.34***

***= $p < 0.001$

control and soil stability. The erosion hotspot maps generated by this study provide a practical decision-support tool for prioritizing soil and water conservation measures, guiding sustainable land-use planning, and supporting climate-resilient management in steep, erosion-prone landscapes.

Acknowledgements
We thank Mizoram University and open-source data providers. Distinguishing this work from prior studies, we present original data and an enhanced RUSLE framework using Google Earth Engine and machine learning specific to Aizawl District, without reusing data from previous publications.

Author contributions
Ahmed Abdallah Adam Mohamed Conceptualization, data curation, formal analysis, investigation, software, methodology, visualization, writing first original draft; Chhakchhuak Lalruatkimi, Khumanthem Babina Devi and Nistha Khanna review and editing; Gyati Yam and Ashish Paul Conceptualization and review and editing; Faisal Ismail Musa Investigation, methodology, validation, writing – reviewing and editing; Om Prakash Tripathi Conceptualization, supervision, investigation, methodology, validation, writing – reviewing and editing. All authors have read and approved the final version of the manuscript for publication.

Funding
The study has not received any specific fund.

Data availability
The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 27 October 2025 / Accepted: 3 March 2026
Published online: 15 March 2026

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